Sign Language Alphabet Recognition

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1. Brief Introduction

This project investigates how a computer can locate a hand and recognize the ASL letter shown in a photograph. We study three complementary changes: targeted training augmentation, a second detection attempt after image preprocessing, and descriptions of hand shape. MediaPipe estimates 21 hand points; a classifier then learns from their geometry (Zhang et al., 2020; Shin et al., 2021). We distinguish detection coverage from correct letter recognition: more accepted hand detections do not automatically mean better classification. Static A–Z images are studied, rather than continuous signing or the movements of J and Z.

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In the landmark-based recognition pipeline used in this study, hand landmark extraction determines which images can provide input to the letter classifier. When the detector returns *no_hand* landmarks, the image cannot be used for feature extraction and subsequent training. Improving detection coverage could therefore increase the number of images available for model training.

In our baseline run on 78,000 ASL (American Sign Language) Alphabet training images covering the letters A-Z, MediaPipe returned *no_hand* detection for 17,598 images (22.56%). A further 151 images produced multiple hand detections, while 60,251 images passed the single-hand acceptance criteria. These counts are reported in Section: 4.3 and can be reproduced from `baseline.csv` in the digital supplement for the `test_all` run. These outcomes motivated an investigation of image preprocessing as a second detection attempt for initially missed images.

We investigate a single application of morphological dilation using a square 3×3 kernel, with erosion and a Canny-edge overlay included as comparison methods.

1. Does targeted data augmentation reduce confusion between certain letters more effectively than general data augmentation?

2. To what extent does applying a 3×3 dilation filter recover visually plausible hand landmarks in ASL Alphabet training images for which MediaPipe initially detects *no_hand*?

3. Do explicit hand-shape categories and joint angles improve on, or replace, Shin's landmark directions?

2. Related works

Shin et al. describe ASL letters using distances and directions between estimated hand points (Shin et al., 2021). MediaPipe estimates 21 such points, including depth and a separate 3D world representation (Zhang et al., 2020). We examine whether explicit descriptions, such as a bent finger or two fingers being spread, add useful information to this representation. We distinguish the effect of adding features from the information a fitted model uses: permutation importance measures the latter and can be affected by related features (Fisher et al., 2019).

3. Methods

3.1. Image Preprocessing

For RQ2, image preprocessing is applied as a second attempt to obtain hand landmarks from images for which MediaPipe initially returns no hand detection. Three transformations are compared: dilation, erosion and a Canny-edge overlay. Each transformation modifies the original image independently, after which the same hand detector is applied again. This comparison measures whether changing the input image allows the detector to recover a previously missed hand.

3.1.1. Dilation

Replaces a pixel value in each color channel with the highest value in its vicinity. This allows light areas to expand and contours to change.

3.1.2. EROSION

It uses the lowest value in that area. This allows light areas to become smaller and dark areas to expand.

3.1.3. CANNY OVERLAY

Zoekt randen op basis van veranderingen in helderheid en voegt de gevonden randen toe aan de oorspronkelijke kleurenfoto. In jouw implementatie worden die randen lichter gemaakt.

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3.2. Hand-shape descriptions and orientation (RQ3)

We compare four building blocks: distances (D), Shin's directions between hand points (S), our own categories such as finger bending and spreading (E), and measured joint angles (J). Distances are always included; switching the other three blocks on or off gives eight combinations. Both SVM and LightGBM receive the same combinations and test images. E and J use estimated world points, whereas D and S use image points. A separate control calculates J from the same points as Shin.

After training, we shuffle one feature group between held-out images and measure the loss in precision and F1. Labels and fitted models remain unchanged. These permutation tests use 13,000 fixed images and five shuffles per group. They measure model dependence, whereas retraining the eight combinations measures the effect of adding or removing information.

Separate orientation checks compare mirror training, normalization to one hand orientation, and G/H rotation. Copies remain in their source image's training fold. Definitions, settings and controls are in Appendix B–E.

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4. Experimental Results

4.1. Datasets

For RQ3, classification uses only the 60,251 original images with an accepted single-hand detection; missing detections are excluded. Five fixed folds separate blocks of consecutive file numbers. These blocks are not verified participants. Every representation uses the same held-out sources, and every image receives one out-of-fold prediction. Recovered images from RQ2 are not added to this feature comparison. Its scores therefore describe classification after accepted detection, rather than the full recognition pipeline.

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For RQ2, we use the A-Z training images from the Kaggle ASL Alphabet dataset. This subset contains 78,000 images, with 3,000 images per letter. The additional classes *space*, *delete*, and *nothing* are outside the scope of this experiment.

All original images are first processed by MediaPipe. Images with the status `no_hand` are selected for the recovery experiment. The supplied Kaggle test images are excluded from RQ2. This experiment evaluates hand-detection recovery within the original training images.

Image N47 (Figure **??**) is used as a development example to inspect the effect of dilation. The complete evaluation includes N47 and images inspected during earlier development. It therefore describes the results on the full dataset, including previously examined images.

4.2. Implementation Details

The feature comparison trains eight representations with two learners across five folds: 80 models. Model settings remain fixed across representations. Only J is imputed and standardized, using the training fold. We report macro-precision and macro-F1 over A–Z, giving each letter equal weight. A paired comparison counts corrected and newly introduced errors on the same images. Appendix B gives the settings, feature dimensions and reproducible result paths.

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MediaPipe Hand Landmarker processes each image in IMAGE mode, with both detection and presence thresholds set to 0.50. Up to two hands are detected so that multiple-hand outputs can be identified. The same model and settings are used for the original and processed images.[1] The software versions, settings and checksums are recorded in **??** `settings.json` in the digital supplement for the `test_all` run.

Each image with the original status `no_hand` is processed using three separate variants. Dilation and erosion each use a square 3×3 kernel with one iteration. Both operations are applied to the colour image at its original resolution.

For the Canny-overlay variant, the image is converted to grayscale and smoothed using a 5×5 Gaussian

[1] The detector configuration and output validation are implemented in `research/data.py`, in the `Detector` class.

filter. Canny edge detection uses thresholds of 50 and 150. The resulting edge map is converted to three channels and added to the original colour image, using weights of 0.25 for the edges and 1.0 for the original image.

Each variant is created independently from the original image. The processed images are saved as lossless PNG files and passed to the same detector during the retry step.[2]

Outputs are paired using the original image identifier. Technical recovery is defined as a transition from `no_hand` to `ok`, requiring exactly one detected hand and successful geometric validation. For the combined result, each image is counted once if at least one variant produces `ok`. A systematic visual assessment of the recovered landmarks remains to be completed.

The exported results are analysed using pandas to calculate recovery counts and percentages per preprocessing variant and per letter. Each recovery percentage uses the number of originally missed images in the corresponding group as its denominator. An image may contribute to multiple variant counts, while the combined recovery count includes each image only once.

Matplotlib is used to visualise the results calculated with pandas. The figures are exported as PNG and PDF files. The underlying plotting data are saved as CSV files, together with source-file checksums and plotting-library versions, to support verification and reproduction.[3]

4.3. Results

RQ3: feature comparison. Table 1 compares all eight descriptions. With Shin as the SVM baseline, our categories lose 22 correct predictions; world-joint angles gain 67, and both additions together lose three. The same-coordinate angle control gains only four. The larger world-angle gain therefore cannot be attributed to the angle formula alone. The small SVM improvement does not recur with LightGBM.

In the models with D/S/E, shuffling thumb flexion produces the largest mean F1 drop among our categories for SVM (0.08 percentage points); for LightGBM it is middle–ring spreading (0.03). This identifies model dependence, rather than a guaranteed benefit from adding a feature.

[2]The transformations are defined in the `variants` function in `research/morphology_pilot.py`. The baseline, preprocessing and retry stages are implemented in `research/recovery_experiment.py`.

[3]The calculations and figures are generated by `experiments/hand_detection_recovery/plot_results.py`.

Additional orientation checks. On original accepted images, mirror training loses 26 correct predictions; right normalization loses 136. Both help on a separate artificial original/mirrored mixture, where mirror training performs better. This does not establish performance for new left-handed users (Appendix D).

For G/H, adding 180-degree vertical-axis training rotations corrects four original-view errors and introduces five. When G/H test landmarks are also rotated, macro-F1 instead rises from 0.8693 to 0.9757. The added geometry is therefore learned. However, two user-labelled live G examples with the hand back visible remain H; a live H is recovered. These four selected examples are error analysis, not a new accuracy estimate (Appendix E).

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In the baseline run, 60,251 of the 78,000 A-Z training images passed the single-hand acceptance criteria. MediaPipe returned *no_hand* detection for 17,598 images (22.56%) and multiple hands for 151 images (0.19%). The complete per-image detection results are available in `baseline.csv` in the digital supplement for the `test_all` run.

For the development example N47, the original image produced *no_hand* detection. After one application of 3×3 dilation, the detector returned one hand that passed geometric validation under the same confidence thresholds. The measured outcomes for this image are recorded in `n47_example.csv`.

All 17,598 initially missed images were subsequently processed using the three variants. Dilation recovered 3,560 images (20.23%), erosion recovered 2,018 images (11.47%), and the Canny overlay recovered 1,391 images (7.90%). At least one variant recovered 4,799 unique images (27.27%). Figure 1 compares the individual recovery rates with the combined result.

The percentages for the individual variants overlap. An image recovered by both dilation and erosion contributes to both variant counts, but only once to the combined result. The three variants produced 6,969 successful image-filter pairs, corresponding to 4,799 distinct recovered images. The individual percentages therefore cannot be added to obtain the combined recovery percentage.

Of the 4,799 recovered images, 3,024 were recovered by exactly one variant, 1,380 by exactly two variants, and 395 by all three variants. These counts explain the difference between the total number of successful image-filter pairs and the number of unique recovered images.

After all retries, 12,799 images had no accepted de-

Table 1. RQ3: pooled classification on 60,251 accepted original images. D: distances; S: directions; E: own categories; J: world-joint measurements. P is macro-precision (%). All variants contain D.

	SVM		LightGBM	
Hand description	P	F1	P	F1
Distances + directions (DS)	97.39	0.9739	96.80	0.9678
Distances only (D)	94.43	0.9438	93.63	0.9355
DS + own categories (DSE)	97.36	0.9735	96.78	0.9676
D + own categories (DE)	94.64	0.9460	93.72	0.9364
DS + world joints (DSJ)	97.54	0.9752	96.71	0.9669
D + world joints (DJ)	95.05	0.9501	93.83	0.9377
DS + categories + joints (DSEJ)	97.39	0.9738	96.78	0.9675
D + categories + joints (DEJ)	94.97	0.9493	93.87	0.9379

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Figure 1. Technical recovery among the 17,598 originally missed images. Individual filter groups overlap because an image can be recovered by multiple variants. The combined result counts each recovered image once. Percentages were calculated using pandas and plotted with Matplotlib.

tection under any variant. Of these, 12,779 images retained the status `no_hand` under all three variants. The remaining 20 images produced another rejection status under at least one variant.

The per-image outcomes and successful variants are recorded in `per_image.csv`. Images without an accepted retry are listed in `unrecovered.csv`. Images with `no_hand` under every variant are listed separately in `still_no_hand.csv`. These results quantify technical recovery. The proportion of visually plausible recoveries has not yet been established.

To assess repeatability, the complete experiment was executed a second time on the same workstation using identical recorded settings, software versions and model files. Both runs produced identical detection outcomes for all 78,000 original images and identical checksums for all 52,794 processed images. The retry statuses also matched for every image and variant. For all 6,969 successful retries, the stored image and world landmark coordinates were exactly equal. Both runs recovered 4,799 unique images, corresponding to 27.27% of the originally missed images. The comparison results and checksums of the compared output files are recorded in **??** **??** in the digital supplement for the `test_all_repeat` run.

4.4. Discussion

RQ3: interpretation and limitations. Our categories do not consistently improve on Shin, and categories plus joint angles do not fully replace his directions under the tested settings. The small world-angle improvement depends on the learner and coordinate source. Permutation dependence should not be read as the gain from adding a feature.

The G/H rotation learns artificial rotated geometry but does not resolve the observed live G/H confusion. A correct label on the original dataset is also not an independent assessment of ASL execution. Real hand-back views can yield different estimated hand points; rigid rotation preserves the original geometry and does not recreate those detection changes.

The folds were inspected during development and are not verified people, so these are exploratory results rather than an untouched final test (Cawley & Talbot, 2010). Detection recovery, accepted-image classification and selected webcam errors have different denominators and should be interpreted separately.

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Dilation produced the highest technical recovery rate of the three evaluated variants. The higher combined recovery rate shows that the other variants also recovered images

that dilation did not recover. However, the combined result requires trying all three variants and accepting an image when at least one attempt succeeds.

The N47 example demonstrates that a small change to the input pixels can change the detection outcome while the model and thresholds remain fixed. The experiment does not establish which individual change in a contour, texture or contrast caused this response. Such explanations require further controlled comparisons and visual inspection.

The results are limited to the evaluated dataset and detector configuration. N47 and earlier development images are included in the complete run, and the reported results therefore include previously examined examples. In addition, technical recovery does not establish the accuracy of the estimated hand points. A visual assessment is needed to determine whether the recovered landmarks are plausible.

The agreement between the two runs supports repeatability under the recorded conditions. Repeatability across other computers, software versions or detector models was not evaluated. The coordinate comparison covers the retry detections, as landmark coordinates from the original baseline images were not retained.

The contribution to RQ2 is therefore the measured technical recovery of initially missed images. Whether these recovered images improve letter recognition requires a separate training and evaluation experiment.

References

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A. Appendices

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B. RQ3: feature definitions and evaluation

This supplement gives the feature definitions, evaluation protocol and separate orientation controls supporting RQ3.

B.1. Source selection, folds and model settings

The feature study uses 60,251 accepted original A–Z images. The original Kaggle test images duplicate training sources and are not used as an independent test. Five frozen folds separate six blocks of consecutive file numbers, not verified participants. Every representation shares source IDs, labels and held-out partitions. Test folds were inspected during development; this is exploratory validation. The historical feature-cache detector thresholds were not retained. Matching image counts do not establish equivalence with the separately recorded RQ2 detector configuration.

Table 2. Feature blocks. D is always retained; S, E and J are independently included or omitted.

Block	Inputs	Meaning
D	190	Normalized distances between image landmarks following Shin.
S	630	Direction cosines for 210 landmark pairs relative to the image axes; these are not internal joint angles.
E	70	One-hot categories for the 19 own checks: flexion, spreading, crossing, thumb shape and contact.
J	15	Fourteen internal angles and one signed thumb deviation from estimated world landmarks. MCP measurements are geometric approximations.

The eight representations are DS, D, DSE, DE, DSJ, DJ, DSEJ and DEJ. DSEJ contains 905 inputs. E is derived from measured geometry, not from the correct letter. Original images are used without augmentation or right normalization in this factorial comparison. D/S/E/J were recomputed on 390 sources; all 19 one-hot groups were checked across the full cache.

SVM uses an RBF kernel, $C = 10$, `gamma=scale` and balanced classes. LightGBM uses 100 trees, 31 leaves, learning rate 0.1, minimum 20 examples per leaf and seed 42 (Ke et al., 2017). Only J receives median imputation and standardization fitted on the training fold. No representation-specific parameter search is performed. Macro-precision and macro-F1 average equally across A–Z, assigning zero to undefined letter scores.

Table 3. Paired changes relative to DS. Net is corrected minus newly introduced errors on the same 60,251 images.

	SVM			LightGBM		
Variant	Corrected	Lost	Net	Corrected	Lost	Net
DS	0	0	+0	0	0	+0
D	295	2,180	-1,885	504	2,524	-2,020
DSE	88	110	-22	116	117	-1
DE	320	2,080	-1,760	495	2,465	-1,970
DSJ	143	76	+67	98	146	-48
DJ	325	1,838	-1,513	511	2,405	-1,894
DSEJ	145	148	-3	133	150	-17
DEJ	341	1,908	-1,567	506	2,389	-1,883

The same-coordinate J control uses Shin's image points without aspect correction and shares sources, folds and model profiles. It gains four SVM predictions, compared with 67 using world J. The coordinate source is therefore a relevant alternative explanation.

B.2. Permutation procedure

Within each fold, seed 20261107 selects 100 held-out images per letter: 2,600 per fold and 13,000 unique sources in total. Five seeded row permutations move all columns in a feature group together, preserving valid one-hot encodings. Labels and trained models remain fixed. D/S/E/J blocks are tested across the eight variants; the 19 E checks are tested in DSE/DE without J, and E/J families in DSEJ/DEJ. The resulting 3,900 interventions retain precision and F1 drops. Spread over folds/repeats is descriptive, not an independent confidence interval. Related features can substitute for one another, and shuffling can create unusual combinations; importance is not a causal feature ranking (scikit-learn developers, n.d.; Fisher et al., 2019).



Figure 2. The 19 own checks, with all one-hot columns of a check shuffled together. Values are mean macro-F1 losses in percentage points over five folds and five shuffles. DSE contains Shin's directions; DE omits them. J is absent in both. Negative losses are retained.

Code and evidence: `research/feature_study.py`, `results/feature_factorial_01/`. The protocol, 80 fold models, predictions, paired changes, controls, permutation outputs and verification preserve all values underlying the main table. The optional optimized SVM replay checks its outputs against ordinary SVC predictions; it does not alter the experiment.

C. Own hand features and the inspection tool

The 19 checks comprise 13 flexion checks, two spreading checks, one crossing check, one thumb-shape check and two thumb-tip proximity checks. Multiple categories produce 70 one-hot inputs. They describe estimated landmarks, not independently verified anatomy.

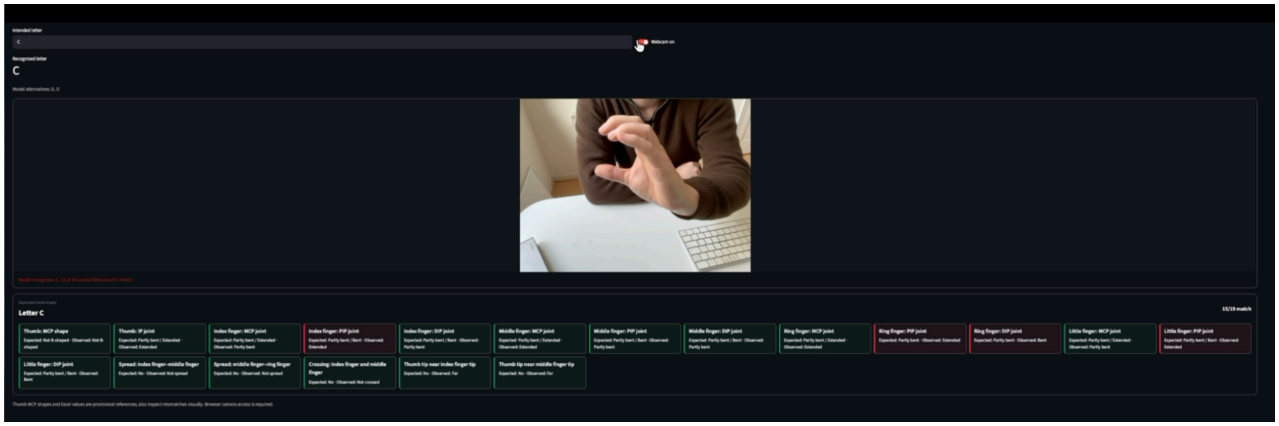


Figure 3. User-supplied feature-tool screenshot. Intended and predicted letter are both C, while only 15 of 19 workbook checks match. The target letter selects expected categories; it does not enter the classifier. Green/red cards indicate agreement with the chosen rules, not confidence in the predicted letter.

Table 4. Own feature ideas and their implemented approximations. Thresholds are project choices, stored in `config/thresholds.json`.

Feature	Measurement and interpretation
Finger flexion (13)	Thumb IP and MCP/PIP/DIP of the other four fingers. A straight joint approaches $180°$. Bent at at most $130°$, extended from $150°$, partly bent between these. Thumb IP uses $80°$ and $155°$.
Spreading (2)	Index–middle and middle–ring evaluated separately: normalized lateral gap at least 0.25, depth gap at most 0.6, positive direction cosine, and PIP/DIP angles at least $120°$.
Crossing (1)	Index and middle reverse lateral order along their finger profiles, with depth difference at most 0.7 and sufficient extension. Crossing lines in 2D alone are insufficient.
Thumb shape (1)	The idea of the thumb across versus beside the palm is approximated by signed thumb-MCP deviation: B-shaped above $16°$ towards the palm. This does not directly measure palm coverage.
Thumb-tip proximity (2)	Separate distances to index and middle tips, divided by wrist–middle-MCP length: contact at most 0.1; near up to 0.825; far above this.

MCP denotes the knuckle, PIP the middle joint, and DIP the end joint; the thumb has an IP joint. The measurements use estimated 3D world points. “Not spread” is not the same as touching: the pair can be adjacent, bent or crossed. Direct contact is only checked for the two thumb-tip pairs. “Not B-shaped” includes several thumb positions.

The workbook can allow several categories per letter. For “no contact”, both near and far count as agreement. Unknown measurements are not confirmed agreement; blank reference fields are not scored. In the screenshot, index PIP is observed extended although partly bent/bent is expected, while index–middle is correctly observed not spread. The original screenshot is retained unchanged. The live tool also preserves the actual image, model hash, features and prediction for error analysis.

D. Separate checks of hand orientation

These checks retain their own 835-input SVM reference (DSJ); they are not extra variants of the eight-way feature comparison. Historical source results remain separate.

D.1. RGB mirrors and detector-Right normalization

Original and horizontally mirrored RGB images are independently detected with IMAGE mode, at most two hands, and detection/presence thresholds 0.5. There are 60,251 accepted originals and 60,201 accepted mirrors. Mirror training adds accepted mirrors of fitting sources while retaining total weight one per source. Original-only preprocessing, numeric gamma and class weights are shared. Sources and their copies remain in the same fold.

Normalization instead uses each detection's own hand label: detector-Left image points receive $x' = 1 - x$, world points $x' = -x$, then the hand label becomes Right. Other coordinates and anatomical indices remain unchanged; features are recomputed before fitting and inference. This reflection is not a palm-to-back rotation. Detector-Right is unchanged; unknown labels are rejected.

The fixed artificial mixture selects half the source images per letter/fold for RGB mirroring, using seed 20261106 before checking detection success. It contains 60,225 accepted images. Within each presentation all models use the same accepted sources; failed detections are excluded. This is not a verified equal mixture of real left- and right-handed users.

Table 5. SVM orientation checks: accepted detections only. P is macro-precision (%); each test presentation has its own denominator.

Test presentation	Training/input	N	P	Macro-F1	Correct
Original	Original training	60,251	97.539	0.97524	58,894
Original	Mirror training	60,251	97.495	0.97493	58,868
Original	Right normalized	60,251	97.322	0.97320	58,758
Reflected RGB	Original training	60,201	26.007	0.19249	14,027
Reflected RGB	Mirror training	60,201	97.293	0.97250	58,676
Reflected RGB	Right normalized	60,201	94.885	0.94802	57,239
Fixed mixture	Original training	60,225	73.566	0.62650	36,452
Fixed mixture	Mirror training	60,225	97.451	0.97421	58,801
Fixed mixture	Right normalized	60,225	96.076	0.96047	57,987

On original images, normalization loses 136 correct predictions relative to no change, and mirror training loses 26. On the artificial mixture, normalization improves on no change but remains below mirror training. Detector labels are not independently verified; mirror pairs share pose, person and background. These results do not establish performance for unseen left-handed people. Evidence: `results/mirror_01/` and `results/handedness_01/`.

D.2. Why G/H rotation is a different hypothesis

For usual ASL G/H, the palm faces inward or sideways (Shield et al., 2022; Vicars, n.d.). Selected Kaggle training examples show the palm towards the camera. This motivates a possible training/use mismatch, not a demonstrated fault in Shin's learning method; all labels and camera viewpoints have not been independently checked.

A 180° camera-y rotation maps wrist-relative coordinates (x, y, z) to $(-x, y, -z)$, with determinant $+1$. Horizontal reflection instead changes only x and has determinant -1. A rotated G/H fitting copy and its original each weigh 0.5; other sources retain weight one. The 4,801 G/H sources yield 65,052 fitting rows in the full-data fit. The rotation preserves distances and intrinsic joint measurements; it produces no new RGB image or detector output.

E. G/H: artificial geometry versus live observations

The first test retains all 60,251 accepted original images, including palm-facing G/H labels. The separate artificial control rotates only the 4,801 held-out G/H descriptions; other letters retain their original predictions. Models are not refitted for this test. Neither test independently validates ASL execution. Recognizing an atypical palm-facing G/H is not the intended live-use goal.

Table 6. G/H rotation checks over all 26 letters. Both presentations use the same 60,251 sources, but the artificial test replaces G/H geometry.

Test descriptions	Training	P (%)	Macro-F1	Correct
Original	Original training	97.5390	0.975244	58,894
Original	G/H rotation training	97.5387	0.975233	58,893
Artificial G/H rotation	Original training	85.0689	0.869284	54,151
Artificial G/H rotation	G/H rotation training	97.5861	0.975715	58,924

Rotation corrects four original-view errors and introduces five (net -1). On the artificial control, it corrects 4,776 errors and loses three (net +4,773); G/H recall reaches 99.47%/99.45%. The added geometry is learned, but this is not real hand-back detection. Evidence: `results/rotation_GH_y180_01/` and `results/gh_occlusion_01/`.

E.1. Exploratory approximation of occlusion

A coarse palm ellipsoid and finger volumes are built around source landmarks and rendered with a depth buffer. MediaPipe accepts 64/64 original photographs but 0/64 renders at either 0 or 180 degrees (0/128 total). This rendering route supplies no training detections; known source points are not substituted for failed detector output.

A separate proxy rotates image/world landmarks and adds reproducible Gaussian coordinate noise only to finger joints hidden behind another estimated hand component. The two fixed standard deviations are 2% and 5% of wrist–middle-MCP length, scaled separately for image/world coordinates. All 835 features are recomputed. Original fitting weights, folds, preprocessing, gamma and class priors remain fixed. Ten new fold models give no extra gain over pure rotation on either the original or clean artificially rotated test. This is assumed point uncertainty, not a measured distribution of MediaPipe occlusion errors. No independent real hand-back evaluation is implied.

E.2. Four user-selected live examples

The original stored landmarks, without redetection, are replayed through both frozen deployment models. All four are accepted detections; model hashes, stored input vectors and recorded predictions are checked. Both hand-back G examples remain H after rotation training. An H changes from Z to H, and a G with the palm visible remains G. Intended letters are user annotations and views are described visually, not expert ASL ground truth. The examples are selected error analysis, not a frequency or accuracy estimate.

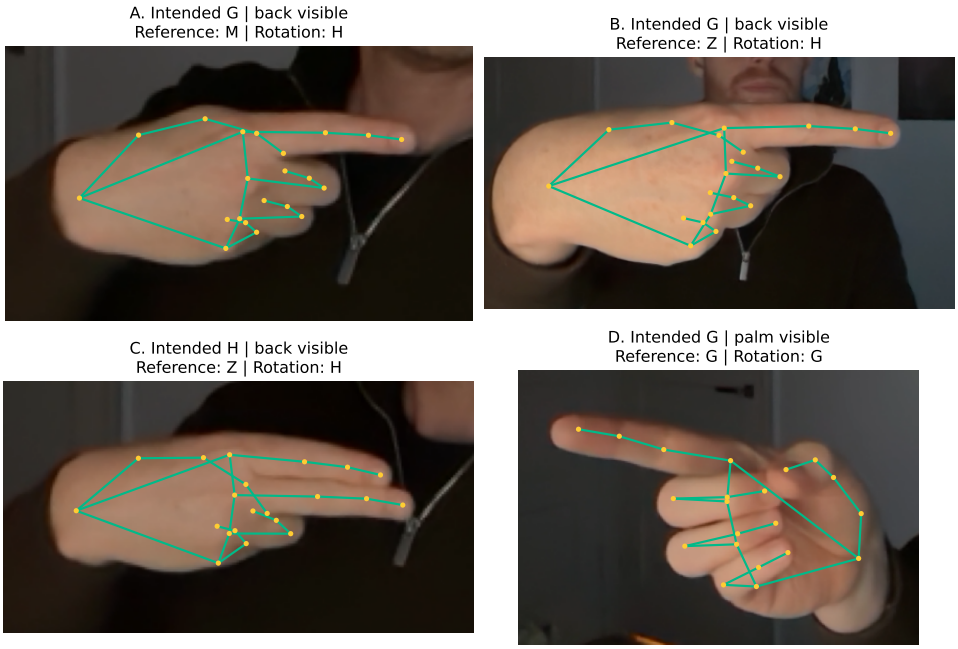


Figure 4. All four new live examples with stored landmarks. A/B: intended G remains H; C: H is recovered; D: G is recognized with the palm visible. "Reference" and "Rotation" are model outputs. No image is used for training.

The middle-finger DIP estimates in A/B are 161.7 and 164.7 degrees, compared with a G-training median of 113.2 (middle 50%: 109.2–118.2). Signed thumb deviation is -1.9 and 2.7 degrees versus median 35.9. These are estimated world-point measurements, not true 3D anatomy. Pose, viewpoint and detector error cannot be separated here, nor is one angle proven to cause the classifier error. A rigid rotation preserves the original joint geometry and does not recreate such new detection changes.

The high artificial score therefore does not directly transfer to the live camera. Rotation training is not yet a demonstrated solution to the observed G/H confusion. This observation does not establish the general failure rate. Source IDs, paired outputs, measurements and hashes: `results/gh_live_diagnostics_20261107/`.